MASTER ENGINEERING PROJECT & NATIONAL DEPLOYMENT BLUEPRINT

PROJECT JAL-SHODHAK (BIO-HARVESTER)

Autonomous Amphibious Canal Restoration Drone with In-Situ Biomass-to-Biofuel Extrusion

DISCIPLINE	MISSION AREA	CORE INNOVATION	OUTPUT VALUE
Mechanical & Mechatronics	Canal Desiltation & Irrigation	Twin-Catamaran Extruder Drone	Clean Waterways + Biofuel Pellets

National Mission Statement: To safeguard the nation's agrarian arteries and rural water security by deploying autonomous amphibious robotic harvesters that clear weed-choked irrigation canals, restore natural drainage to prevent seasonal floods, and convert noxious aquatic biomass into decentralized, high-calorific clean biofuel pellets for rural communities.

1 The National Crisis & Problem Statement

Millions of kilometers of irrigation canals, river tributaries, and municipal water channels across the nation suffer from severe biological and hydrological choke points that jeopardize agricultural output and public safety:

Core Threats to the Nation:

- **Canal Choking by Water Hyacinth (*Eichhornia crassipes*):** Hyacinth weed doubles in biomass every 10–12 days, forming dense floating mats that reduce water velocity and canal throughput by **up to 70%**, causing downstream tail-end farms to dry up during critical sowing cycles.
- **Devastating Monsoon Flooding:** Weed and silt choked drainage canals create massive backflow surges during heavy rains, causing earthen embankments to breach and triggering catastrophic rural and urban flash floods.
- **Ecological Suffocation:** Dense canopy blocks sunlight and consumes dissolved oxygen, triggering mass fish mortality and turning vital water reservoirs into stagnant breeding grounds for disease vectors.
- **Failure of Existing Practices:** Traditional diesel excavators (JCBs) operating from canal banks tear down fragile earthen bunds, burn ~14 L/hr of diesel, and dump rotting weed piles along roadsides that emit foul methane odors and toxic runoff.

🚨 Conventional Backhoe Dredging (Current Method)

Heavy machinery operating from banks. Destroys earthen bunds, burns expensive diesel (~$18/hr), leaves roots intact in the water, and creates foul roadside dumps.

💡 The Jal-Shodhak Solution (Our Innovation)

Ultralight amphibious catamaran entering the water directly. Solar-electric powered, cuts roots 0.8m deep, shreds and converts wet weed into solid biofuel pellets on-board.

2 The 4-Stage In-Situ Circular Processing Pipeline

Rather than merely cutting weeds and transporting wet rot, Jal-Shodhak processes raw aquatic weeds continuously on-board through four integrated stages:

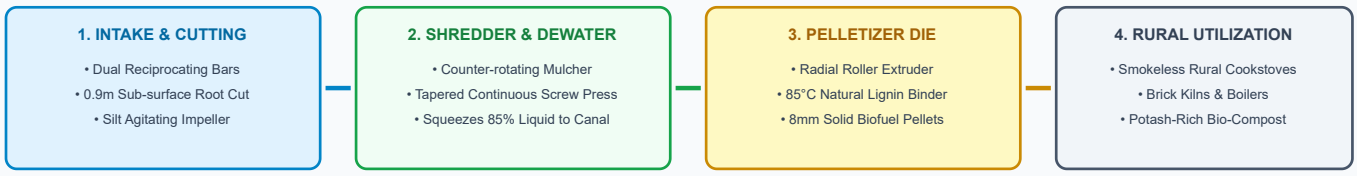


Figure 1: On-board continuous processing pipeline converting raw aquatic weeds into high-density energy pellets.

3 Mechanical & Mechatronic Engineering Subsystems

A. Amphibious Twin-Catamaran Hull & Crawlers

Constructed from marine-grade corrosion-resistant **Aluminum 5083-H116** and rotomolded HDPE floats. Features an ultra-shallow operational draft of **0.22 meters** to glide over shallow weed beds and mud shoals.

- **Propulsion:** Dual 48V steerable BLDC Kort-nozzle thrusters with weed-chopping hub blades to prevent fouling.
- **Bank Traversal:** Retractable paddle-crawler tracks allow the drone to climb across canal bunds, locks, and weirs.

B. Submerged Harvester & Mulcher

Front articulated conveyor table equipped with dual reciprocating cutter bars engineered with **tungsten-carbide serrated teeth**. Submerges up to 0.9 m to cleanly slice root crowns.

- **Hexagonal Shredder:** High-torque twin shafts running at 180 RPM shred dense vegetation into 15–20 mm uniform mulch.
- **Anti-Jam Torque Monitor:** Dynamic load sensing automatically reverses cutters if nylon rope or scrap debris is detected.

C. Continuous Screw-Press Dewaterer

Fresh water hyacinth contains 92–95% water by weight. Jal-Shodhak uses a progressive tapered compression screw with a slotted wedge-wire cylindrical screen.

- **Moisture Drop:** Reduces moisture from 94% down to under 35% in seconds.
- **Clean Effluent Return:** Filtered, oxygenated water returns to the canal, replenishing local dissolved oxygen.

D. Radial Die Extrusion Pelletizer

The compressed biomass cake enters a rotary roller die extruder. Motor cooling jacket heat (85°C) liquefies the plant's natural lignin, bonding dense 8mm solid fuel pellets without chemical additives.

- **Energy Density:** ~16.8 MJ/kg (competes directly with dry wood briquettes).
- **On-Board Bunker:** 350 kg capacity with automated offloading auger for seamless canal-bank unloading.

4 Mechatronics, Sensors & Autonomous Navigation

The drone operates autonomously via centimeter-accurate satellite positioning and onboard Edge AI computer vision:

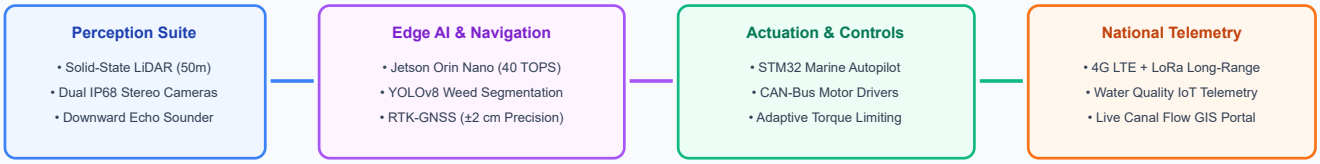


Figure 2: Mechatronics sensor-to-actuation architecture with real-time cloud GIS synchronization.

5 Prototype Bill of Materials (BOM) & Budget

Subsystem	Component Specification	Qty	Key Technical Metric	Est. Cost ($)
Hull & Chassis	Al-5083 Catamaran Hull + Rotomolded HDPE Floats	1 Unit	Displacement: 950 kg; Draft: 0.22 m	$1,850
Propulsion	Dual 48V 1.8kW BLDC Thrusters with Kort Nozzles	2 Units	Bollard Thrust: 65 kgf; Max Speed: 6 knots	$920
Cutter Bar	Reciprocating Tungsten-tipped Cutter Bar	1 Set	Cut Width: 1.6 m; Depth: 0.9 m; 1200 rpm	$780
Shredder/Press	Twin-Shaft Shredder + Tapered Screw Dewaterer	1 Unit	Torque: 420 Nm; Moisture Drop: 94% → 32%	$1,450
Pelletizer Die	Continuous Radial Die Pellet Mill (4kW BLDC Core)	1 Unit	Pellet Diam: 8mm; Throughput: 120 kg/hr dry	$1,100
Solar Canopy	High-Efficiency Monocrystalline Bifacial Panels	4x 320W	Total 1.28 kWp; MPPT Charging Controller	$640
Battery Pack	LiFePO4 48V 120Ah Deep-Cycle Power Storage	2 Packs	11.5 kWh total capacity; 3500+ lifecycle	$1,600
Compute & Vision	NVIDIA Jetson Orin Nano + IP68 Stereo Depth Cams	1 Set	YOLOv8 Edge Model for Weed / Trash Classification	$750
Navigation/IoT	Dual-Antenna RTK-GNSS + 4G LTE/LoRa Gateway	1 Set	Centimeter Positioning + Cloud Irrigation Portal	$420
TOTAL PROTOTYPE DEVELOPMENT BUDGET (Phase-1 Scale Unit)			**Ready for Field Trials**	**$9,510**

6 Comparative Advantage Over Conventional Methods

Comparison Parameter	Conventional JCB / Excavator	Imported Weed Harvester Boats	Jal-Shodhak Bio-Harvester
Capital Cost (Capex)	$55,000 – $80,000	$150,000 – $320,000	**$9,500 – $14,000** 85% Lower
Operational Energy	Diesel (~14 L/hr = $18/hr)	Diesel (~22 L/hr = $28/hr)	**Solar + LiFePO4 ($0.40/hr)** 98% Cheaper
Canal Bund Impact	Severe erosion & road collapse	Requires wide docks; damages bunds	**Zero bank impact** (Amphibious float)
Biomass Disposition	Rotting heaps on roads; toxic runoff	Dumped on barges; double transport	**In-situ pelletization** (Valuable fuel)
Carbon Footprint	Net Emitting (~38 kg CO2 / hour)	Net Emitting (~58 kg CO2 / hour)	**Net Carbon Negative** (Biofuel offset)

Circular Economy Revenue Model: A single drone clearing 1.2 hectares per 8-hour shift produces approximately **1,800 kg of dense dry biofuel pellets daily**. Sold to local rural brick kilns or tea factories at $0.08/kg, the drone generates **~$144/day in clean green revenue**, fully amortizing its capital construction cost in less than 4 months of operation!

7 Field Deployment SOP: Pre-Float & Software Boot

Follow this exact Standard Operating Procedure (SOP) before putting the autonomous drone in the water:

Phase 1: Site Survey & RTK Setup

Fly a low-altitude mapping drone or use Sentinel-2 satellite NDVI data to locate weed choke-points. Set up the RTK-GNSS base station tripod on a firm canal embankment point. Confirm LoRa telemetry differential lock (±2cm accuracy).

Phase 2: Water Entry Slipway Setup

Back the custom low-bed transport trailer into the canal ramp or slipway. Release the electric winch strap. The drone floats off into the water, engages reverse Kort thrusters, and holds station via dynamic positioning.

Pre-Float Hardware Inspection Checklist

Check Item	Inspection Procedure	Expected Threshold	Action if Non-Compliant
Hull Watertightness	Inspect pontoon hatches, O-rings, and cable glands	Zero moisture in bilge compartments	Replace silicone gasket & vacuum test
LiFePO4 Battery Pack	Check BMS telemetry on smart display	Voltage > 51.2V (State of Charge > 90%)	Connect solar fast-charger or hot-swap
Cutter Bar & Shredder	Verify blade clearance and lubricate drive eccentrics	Free rotation; cutter gap < 0.4 mm	Re-shim blade guide and tighten drive clamp
Screw Press & Die	Inspect slotted wedge-wire screen for residual fibers	Clear extrusion channels; pre-heat die to 60°C	Run manual cleanout cycle using water jet
Emergency Kill-Switch	Test physical latch switch and remote 433MHz RF beacon	Immediate thruster and motor cutoff (<100ms)	Do not launch until relay safety is verified

Software Stack Boot Sequence (NVIDIA Jetson Orin Nano)

```
$ sudo systemctl start jalshodhak-marine-core.service   # Launches ROS2 core & CAN-Bus motor driver
$ ros2 launch perception stereo_lidar_fusion.launch.py      # Initializes IP68 cameras & solid-state LiDAR
$ ros2 run weed_detector yolov8_waterweed_tensorrt          # Loads 40 TOPS accelerated weed segmentation
$ mavproxy.py --master=/dev/ttyTHS1 --baudrate 921600        # Opens telemetry bridge to STM32 ArduPilot
```

Mission Upload via QGroundControl / Mission Planner

Define the canal clearing boundary polygon. Set the survey pattern with 1.4m lane overlap. Set forward speed to 1.8 knots (0.9 m/s) and submerge the cutter table to 0.7m depth. Upload waypoints over encrypted MAVLink telemetry.

8 Autonomous Harvesting & Biomass Offloading Operations

Step A: In-Situ Cutting & Mulching

The drone engages the submerged reciprocating cutter blades, slicing through root crowns. The intake conveyor continuously sweeps the biomass into the counter-rotating shredder, mulching 120 kg of wet hyacinth per minute into uniform 15mm pulp.

Step B: Continuous Dewatering & Pelletizing

The progressive tapered screw press squeezes out 85% of plant water, returning clear aerated liquid to the stream. The dried fibrous cake is extruded through the heated radial die into 8mm fuel pellets and deposited into the 350kg onboard bunker.

Step C: Precision Docking & Offloading

When the internal pellet hopper reaches 90% capacity, the drone automatically navigates to the canal-bank transfer station. A laser alignment sensor mates with the mobile hopper, discharging dry pellets via an electric auger in 4 minutes.

Step D: Post-Shift Maintenance Flush

Perform a 5-minute high-pressure freshwater wash of the cutter bar and screw press. Inspect propeller Kort nozzles for entangling fishing line or plastic waste. Plug into the canal-side solar charging bank.

9 12-Month Prototype & Field Trial Roadmap

Quarter 1: CAD & Lab Bench Prototyping

- Hydrodynamic simulation (ANSYS Fluent) of catamaran hull under weed drag.
- Fabrication of scale screw-press dewaterer; torque and dewatering test bench.
- Dataset collection and training of YOLOv8 weed-segmentation model.

Quarter 2: Full-Scale Fabrication

- Welding and CNC fabrication of Al-5083 catamaran hull and conveyor structure.
- Integration of 48V powertrain, solar canopy, and LiFePO4 battery management.
- Static buoyancy, balance, and emergency bilge pump validation in tank test.

Quarter 3: Irrigation Canal Field Trials

- Deployment in a 5 km heavily choked agricultural canal pilot stretch.
- Validation of automated RTK-GNSS waypoint navigation and collision avoidance.
- Chemical calorific testing (bomb calorimeter) of extruded biofuel pellets.

Quarter 4: Autonomous Fleet Swarm & Handover

- Implementation of multi-drone swarm coordination for broad lake clearing.
- Creation of open-source maintenance manual and training for rural youth cooperatives.
- Formal presentation to National Irrigation and Water Resources Ministries.

10 National Institutional Strategy: Where & How to Deploy

To scale this from a working prototype to a nationwide sovereign water rejuvenation program:

1. Target Government Nodal Agencies (Where)

State Water Resources / Minor Irrigation Dept: Mandated with canal desiltation.
Municipal Corporations & Smart Cities: Lake rejuvenation & storm drainage.
National River Rejuvenation Missions: Namami Gange, State Wetland Authorities.

2. The 5-km Demonstration Pilot Corridor (How)

Submit proposal to the **Executive Engineer (Irrigation)** or District Collectorate for a zero-cost 5 km trial. Present before-and-after satellite NDVI weed coverage, canal flow velocity (cusecs), and water dissolved oxygen data.

3. National Innovation Grants & CSR Capital

Government Grants: Startup India Seed Fund, DST-NIDHI, BIRAC Clean Tech.
PSU CSR Funds: NTPC, NHPC, PowerGrid, and Oil PSUs have mandatory CSR allocations for clean water and green rural livelihoods.

4. Rural Village Cooperative Model (FPOs/SHGs)

License fleet management to local **Farmer Producer Organizations (FPOs)** and **Self-Help Groups (SHGs)**. They operate the drones to protect their crops and sell biofuel pellets ($140/day) as a sustainable village enterprise.

11 Direct National Benefits & Sovereign Impact

Food Security & Irrigation Flow

Guarantees full hydraulic flow through irrigation channels, ensuring water reaches tail-end farmers during critical sowing cycles.

Flood Mitigation & Disaster Resilience

Pre-monsoon canal desiltation prevents backflow surges and structural embankment breaches that cause catastrophic loss of life.

Rural Clean Fuel Independence

Supplies rural homes with smokeless biomass pellets, replacing hazardous firewood burning and reducing acute respiratory illness.

IN Atmanirbhar Technological Sovereignty

Replaces multi-crore imported diesel dredger boats with indigenous, open mechatronic technology built, serviced, and operated locally.

Conclusion & Call to Action: *Project Jal-Shodhak* proves that mechanical engineering, robotics, and circular economy design can directly solve one of the nation's most persistent ecological and agrarian crises. By transforming an environmental hazard into clean energy and water security, this project serves both the nation and its people.

Project Submitted By: Engineering Research & Innovation Initiative
Department: Mechanical, Mechatronics & Autonomous Systems

Verification Status: Prototype Stage Ready
Documentation: Complete Master Project Blueprint (6 Pages)